# Ngoc Minh NGUYEN

Quantitative Risk Analyst · Bordeaux, France
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## Education

**Master 2 Advanced Studies and Research in Finance - Research Track (M.Sc. in Finance)** 2023–2024
IGR-IAE — Rennes, France
**Ranked first in class, graduated with the mention Très Bien (highest honors).**
Coursework in quantitative finance, risk management, and financial econometrics. Thesis on Machine Learning in French stock markets under the supervision of Mr. Souleymane LAMINOU ABDOU.

**Msc of Engineering in Applied Mathematics** 2019–2024
INSA de Rennes — Rennes, France
Coursework in applied mathematics, numerical methods, and optimization. Thesis on the application of Applied Mathematics in Model Risk Management at BNP Paribas.

**Exchange Program in Mathematics Department** Feb 2023 – Jun 2023
Korean Advanced Institute of Science and Technology (KAIST) — Daejeon, South Korea
Coursework in Business Intelligence, Data Science, and Bayesian Statistics. GPA: 90.33/100.

**Baccalauréat, Mathematics Option** 2016–2019
High School for Gifted Students, Hanoi University of Education — Hanoi, Vietnam

## Experience

**Quantitative Risk Analyst** Nov 2024 – Present
BNP Paribas · Risk Architecture — Bordeaux, France
Lead the monthly IFRS 9 provisions reporting, ensuring reliability and on-time delivery.
Analyze the Retail portfolio (exposure above €100M), identifying the main risk concentrations.
Run impact simulations on EAD, LGD, and PD.
Optimize existing reporting tools, cutting the reporting process from several days to 2 hours.
Study the risk factors driving PD changes across customer segments.
Tech stack: SAS, Python, VBA, Power BI.

**Model Risk Analyst** May 2024 – Oct 2024
BNP Paribas — Bordeaux, France
Developed a Model Risk Rating (MRR) model used to rank the risk of more than 100 models: data collection, machine learning, hyper-parameter tuning, and continuous optimization.

**Data Scientist** Jun 2023 – Sep 2023
NexGuard (now part of NAGRA) — Rennes, France

## Skills

| | |
|---|---|
| **Programming** | Python (pandas, scikit-learn, Streamlit), SAS, VBA, JavaScript |
| **Quantitative methods** | Credit risk (PD, LGD, EAD), IFRS 9 provisioning, model risk, VaR / expected shortfall, Monte Carlo simulation, stochastic calculus, machine learning, credit scorecards (WOE / IV) |
| **Tools** | Power BI, Dataiku, Bloomberg, Git / GitHub Actions |
| **Languages** | Vietnamese (native), English (full professional), French (full professional) |

## Projects

| SYM | NAME | TYPE |
|---|---|---|
| SENT | **Financial Sentiment Dashboard**<br>Market data + financial news, scored locally with VADER & FinBERT — no API cost. | NLP / Markets |
| RISK | **PEA Risk Lab**<br>Market data + risk management for French equities and UCITS ETFs. | Risk / Portfolio Management |
| SIM | **Strategy Simulation**<br>Market data + strategy testing — DCA, momentum, and Monte Carlo backtests. | Strategy / Backtesting |
| PULSE | **France Economic Pulse**<br>Open-data nowcast of French industrial production, refit and published daily. | Macro / Nowcasting |
| DIST | **Distribution Lab**<br>16 probability distributions to explore, fit to data by MLE, and run through Monte Carlo VaR scenarios. | Probability / Statistics |
| SDE | **Visual Stochastic Calculus**<br>From Brownian motion to Black–Scholes, Heston, and diffusion models — 11 interactive chapters with runnable Python. | Stochastic Calculus / Learning |
| PD | **PD Credit Scorecard**<br>WOE-binned logistic scorecard on 150k Kaggle loans — test AUC 0.85, Gini 0.70. | Credit Risk / Scoring |

## Certifications

**SAS Programming 1: Essentials** — Dec 2024
SAS

**Core Designer Certificate** — Jul 2024
Dataiku Academy

**Bloomberg Market Concepts** — Nov 2023
Bloomberg for Education

**Neural Networks and Deep Learning** — Apr 2023
DeepLearning.AI · Coursera

**AMF Certification**
Autorité des marchés financiers (AMF)

## Competitions

**Cod'INSA** — 2024
Programming competition between the schools of the INSA Group in France

**Hanoi Open Mathematics Competition (HOMC)** — 2015
First Prize